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Optical properties of selected native and coagulated human brain tissues *in vitro* in the visible and near infrared spectral range

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Abstract

Medical laser applications require knowledge about the optical properties of target tissue. In this study, the optical properties of selected native and coagulated human brain structures were determined *in vitro* in the spectral range between 360 and 1100 nm. The tissues investigated included white brain matter, grey brain matter, cerebellum and brainstem tissues (pons, thalamus). In addition, the optical properties of two human tumours (meningioma, astrocytoma WHO grade II) were determined. Diffuse reflectance, total transmittance and collimated transmittance of the samples were measured using an integrating-sphere technique. From these experimental data, the absorption coefficients, the scattering coefficients and the anisotropy factors of the samples were determined employing an inverse Monte Carlo technique. The tissues investigated differed from each other predominantly in their scattering properties. Thermal coagulation reduced the optical penetration depth substantially. The highest penetration depths for all tissues investigated were found in the wavelength range between 1000 and 1100 nm. A comparison with data from the literature revealed the importance of the employed tissue preparation technique and the impact of the theoretical model used to extract the optical coefficients from the measured quantities.

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1. Introduction

Rapid development of laser medicine (Richards-Kortum and Sevick-Muraca 1996, Schuitmaker *et al* 1996, Castro *et al* 1992) has stimulated the investigation of optical properties of different biological structures (Marchesini *et al* 1989, Peters *et al* 1990, van der Zee *et al* 1993, Yaroslavsky *et al* 1996b, Taddeucci *et al* 1996) since the efficacy of laser procedure depends on the photon propagation and fluence rate distribution within irradiated tissues.

In brain applications, substantial efforts have been devoted to the development of optical diffuse tomography, which is a promising monitoring and imaging modality (Chance and Alfano 1995). This technique offers the possibility to obtain physiologically relevant information about the oxygen saturation of brain tissues. Quantitative information, however, can be retrieved only if the optical properties of the brain tissues are known. Considering the therapeutic procedures, laser-induced interstitial heating of deep brain tumours is a promising minimally invasive technique (Amin *et al* 1993, Kahn *et al* 1994, Muschter and Hofstetter 1995, Hata *et al* 1998, Reimer *et al* 1998) for the treatment of neoplastic lesions. For successful therapy planning and dosimetry, a database of the optical properties of tumours and of surrounding native brain tissue is needed. During laser therapy, the optical characteristics may change altering the penetration depth of the irradiated tissue significantly (Agah *et al* 1994). Therefore, heat-induced changes of the optical properties in the tissue should be determined. In addition, knowledge of brain optical properties allows identification of the specific wavelength range where light penetration depth is maximal. These wavelengths are optimal for laser thermotherapy treatment.

Optical constants of native white and grey brain substance may be found in the literature for selected wavelengths. Data for coagulated human brain tissue are restricted to white brain matter and are available for few wavelengths only. The optical properties of brainstem structures (e.g. thalamus, pons) and cerebellum, to the best of our knowledge, are unavailable so far. Therefore, the aim of this study was to determine and compare the optical properties of native and coagulated brain tissue from a variety of regions *in vitro* in a practically relevant spectral range from 360 nm to 1100 nm using integrating-sphere measurements combined with an inverse Monte Carlo technique.

2. Materials and methods

2.1. Sample preparation

Specimens were obtained from seven non-diseased human brains during autopsy performed less than 48 h after death. The material was stored at low temperature (2–3 °C, <48 h) until spectroscopic investigation. From these specimens, 14 samples (2 × 2 × 2 cm) of each brain structure were prepared. To simulate the laser coagulation process, seven of these 14 tissue samples were heated in a saline bath for 2 h at 80 °C. The remaining seven samples were left untreated. Sections for spectroscopic measurements were cut using a microcryotome. Cryosections of 80–150 μm thickness were cut from both native and coagulated white matter samples. Cryosections of all other normal tissues samples, native and coagulated, were 100–200 μm thick. Cryosections were rinsed in saline solution (pH 7.4) to remove haemoglobin. Each section was positioned between a microscope slide and a cover slip. The gap at the edges was sealed with an embedding medium for histological sections (Entellan™, Merck, Darmstadt, Germany) to prevent water loss of the tissue. The thickness of the samples was controlled using a micrometer. A total of seven untreated and seven coagulated samples of each brain tissue type (i.e. white matter, grey matter, thalamus, pons, cerebellum) was investigated. In addition, tumour tissues (meningeoma ($n = 6$), astrocytoma WHO grade II ($n = 4$)) were obtained from patients during surgery. The tumour tissue was stored in isotonic saline solutions at low temperature (2–3 °C, ~2 h) until preparation. Samples with thickness of approximately 300 μm were prepared in the same way as described above. Lateral dimensions of tumour sections were not smaller than 1.5 cm × 1.5 cm. Only untreated samples were measured because the tumours were too small to investigate both the untreated and the coagulated state. To correlate alteration of the optical properties with structural changes of brain tissues induced by coagulation, samples from all specimens were fixed by immersion in 4% phosphate-buffered formalin and embedded in paraffin. Six-micrometre-thick sections were cut serially from each block. These sections were stained with H&E and Masson's trichrome for the histological investigation.

2.2. Integrating-sphere measurements

Spectroscopic measurements were performed as described earlier (Schwarzmaier *et al* 1995). A schematic of the experimental arrangements is presented in figures 1(a)–(c). In short, a xenon lamp (XBO 150 W1, Osram, Germany) combined with a prism monochromator (MQ3, Zeiss, Oberkochem, Germany) served as a monochromatic light source. The output slit width of a monochromator was adjusted to obtain a spectral bandwidth (FWHM) of approximately 10 nm within the spectral range of interest. The beam size at the sample was 2 mm in diameter. For diffuse reflectance and total transmittance measurements (figures 1(a) and (b)) a Spectralon^{TM} integrating sphere was used (IS-060-SL, Labsphere, USA). To determine the collimated transmission, two additional pinholes were employed to limit the acceptance angle of the detector to 10^{-4} sr (figure 1(c)). The reflected and the transmitted light was detected by a silica photodiode with a built-in preamplifier (HUV-2000 B, EG&G, Canada). The signals from the detector were converted to a digital form and stored in a computer. In all experiments, collimated irradiation of the samples was used because in this case the optical properties of the investigated tissue could be defined more accurately than in the case of diffuse irradiation (Pickering *et al* 1992). A single-integrating-sphere configuration was chosen for our experiments to prevent the light interaction between spheres occurring in double-integrating-sphere setting (Pickering *et al* 1993). During the first measurement of each sample, the illuminated tissue area was marked on the glass slide to ensure the irradiation of the same spot of the particular sample during all measurements.

2.3. Data processing technique

For data processing, we used the inverse Monte Carlo technique described earlier (Yaroslavsky *et al* 1996d). The technique employs a quasi-Newton inverse algorithm (Dennis and Schnabel 1983) combined with a forward Monte Carlo simulation. The quasi-Newton inverse algorithm features a fast local convergence and the ability to reach proximity of the solution even from a poor initial approximation. The Monte Carlo algorithm takes into account exact optical and geometrical configuration of the experiments, i.e. mismatched sample–glass and glass–air boundary conditions, losses of light at the edges of the sample, finite sizes of the incident beam, finite port dimensions of the integrating spheres, arbitrary angular distribution of the incident light and the contribution of scattered light into the measured collimated signal.

To obtain correct values of the optical properties from the measured quantities, the experimental data of the diffuse reflectance and the total transmittance were corrected for the difference in sphere efficiency between the sample and the reference probe measurements (Pickering *et al* 1992). From the corrected values of total transmittance, diffuse reflectance

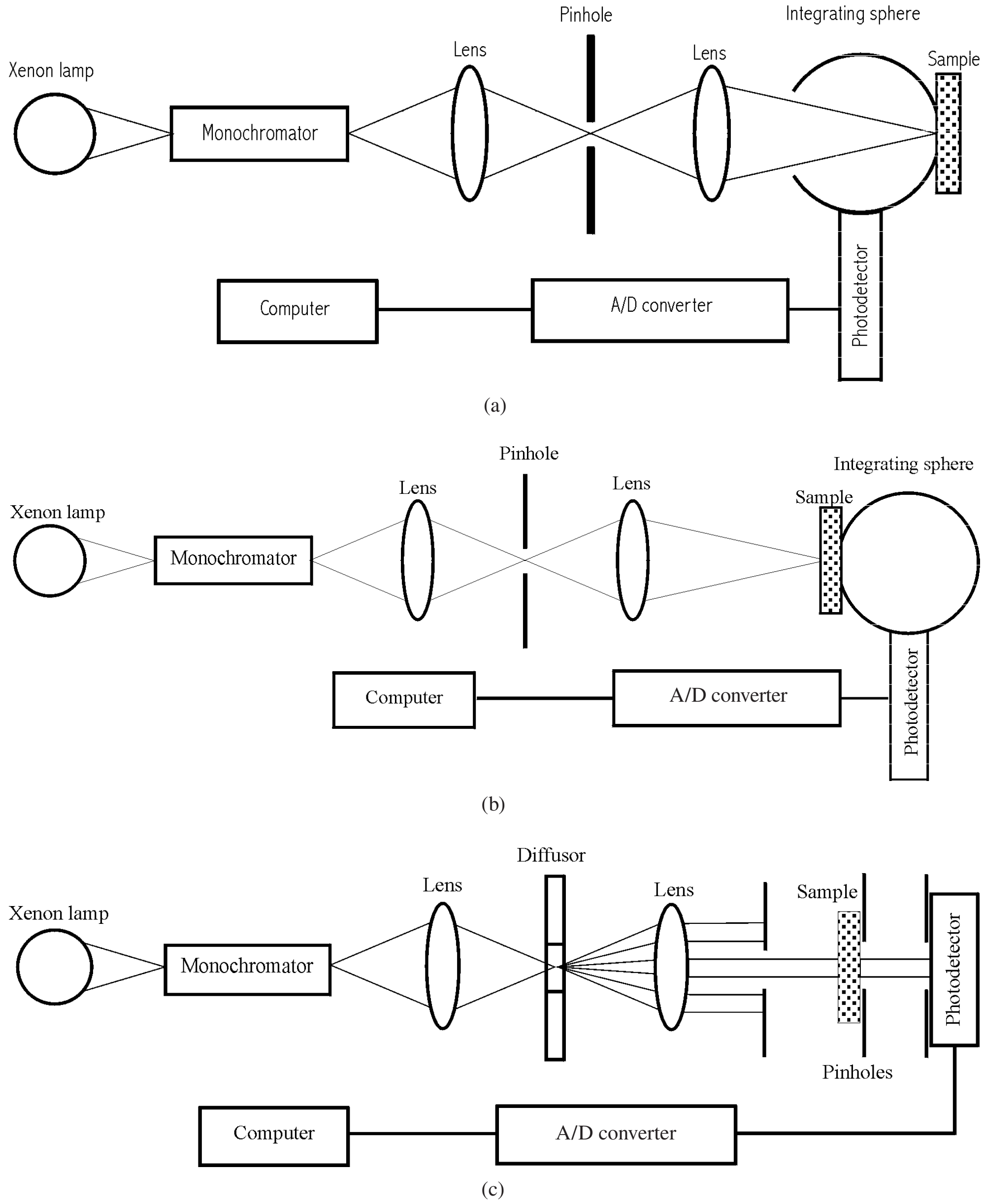
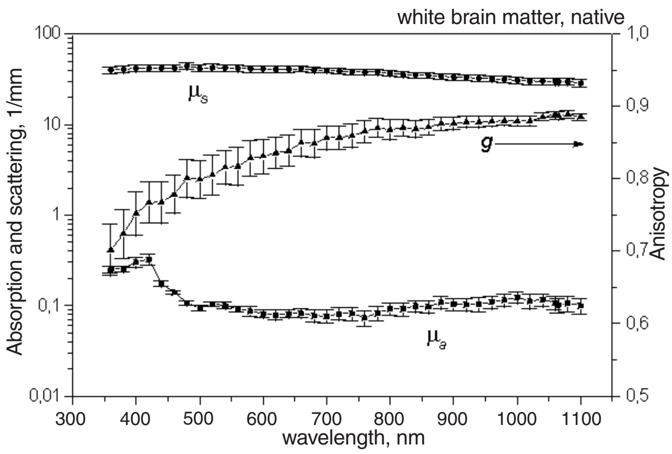
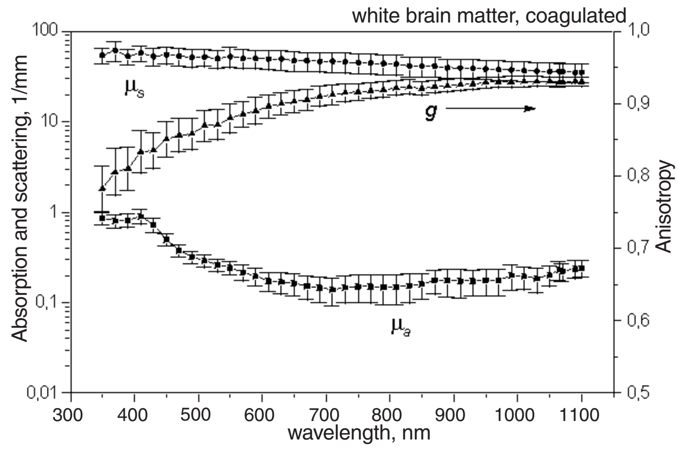


Figure 1. Experimental configurations for (a) diffuse reflectance measurement, (b) total transmittance measurement and (c) collimated transmittance measurement.

and collimated transmittance, the absorption coefficient μ_a, the scattering coefficient μ_s and the anisotropy factor g were determined. The interpretation of all experimental data reported in this study was conducted under the assumption of Henyey–Greenstein scattering phase function because it has been shown previously (Yaroslavsky *et al* 1996c) that this is a valid phase function approximation for brain tissues. In addition to the values of absorption coefficients, scattering coefficients and anisotropy factors, the values of reduced scattering



(a)



(b)

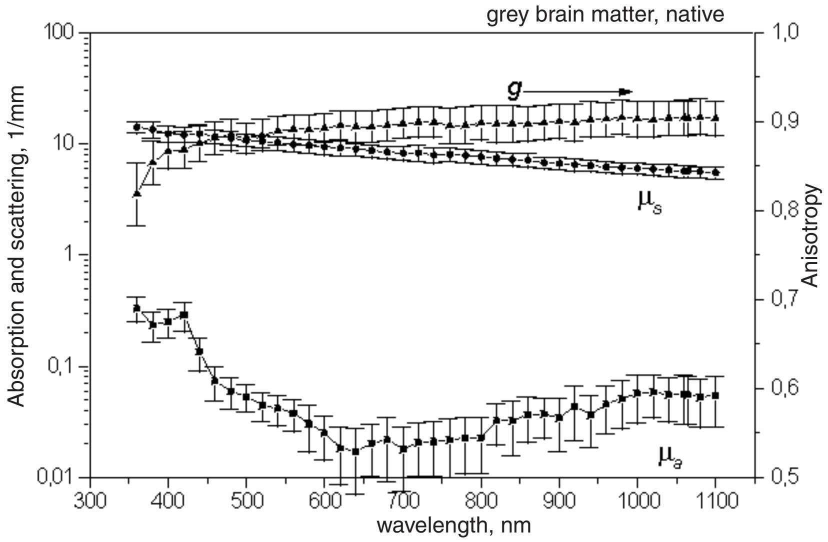
Figure 2. Optical properties of human white brain matter. Average of seven samples. Squares: absorption coefficient, circles: scattering coefficient, triangles: anisotropy factors and bars: standard errors. (a) Native samples and (b) coagulated samples.

coefficients, $\mu_s' = \mu_s(1 - \bar{\mu})$, and of light penetration depth, $\delta_{\text{eff}} = 1/\sqrt{3\mu_a(\mu_s' + \mu_a)}$, were evaluated for selected wavelengths.

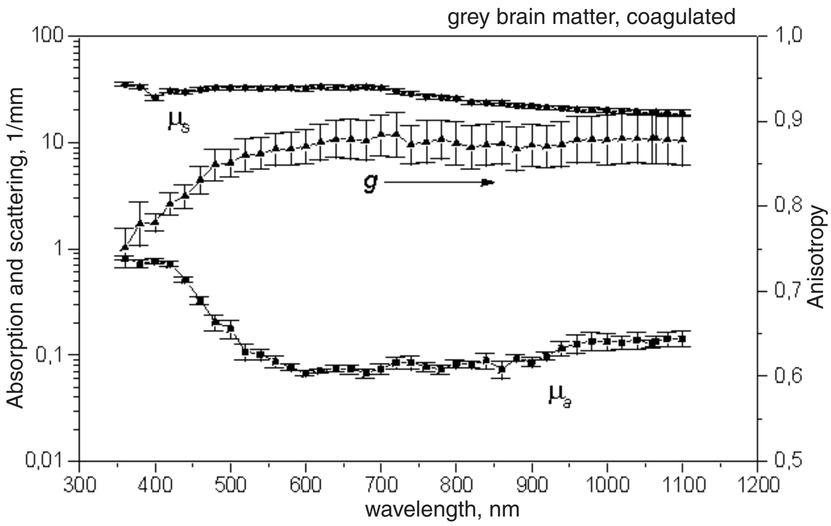
3. Results and discussion

We have investigated seven native and seven coagulated samples of white matter, grey matter, thalamus, cerebellum and pons. In addition, the optical properties of native tumour tissues (meningioma ($n = 8$), astrocytoma WHO grade II ($n = 4$)) were obtained. All samples were measured in the wavelength range from 360 nm to 1100 nm. The results are summarized in figures 2–7.

All brain tissue samples shared qualitatively similar dependencies of the optical properties on the wavelength. The scattering coefficient decreased and the anisotropy factor increased with the wavelength. These findings may be explained by the fact that the contribution of



(a)

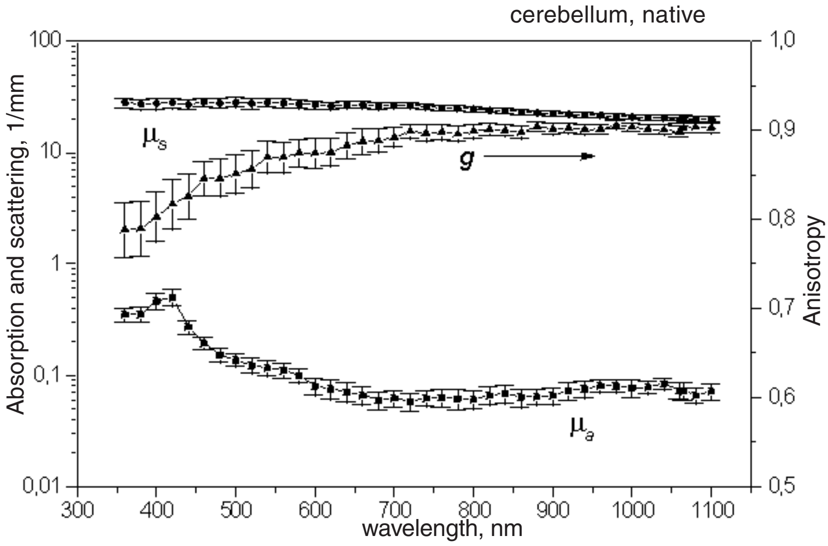


(b)

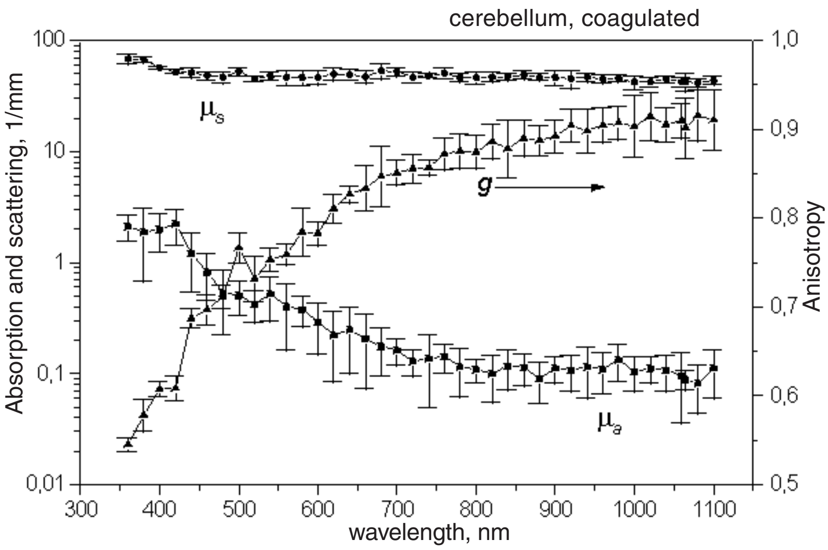
Figure 3. Optical properties of human grey brain matter. Average of seven samples. Squares: absorption coefficient, circles: scattering coefficient, triangles: anisotropy factors and bars: standard errors. (a) Native samples and (b) coagulated samples.

Rayleigh scattering decreases, while the contribution of Mie scattering increases with increase of the wavelength. The wavelength-dependent absorption behaviour of all tissues resembled a mixture of oxy- and deoxy-haemoglobin absorption spectra. Thus, although the samples were rinsed carefully in saline solution, it was obviously not possible to remove all blood residuals from the sections.

At the same time, clear differences in the spectral characteristics of the investigated brain tissues have been observed. For example, the extinction coefficients of white brain



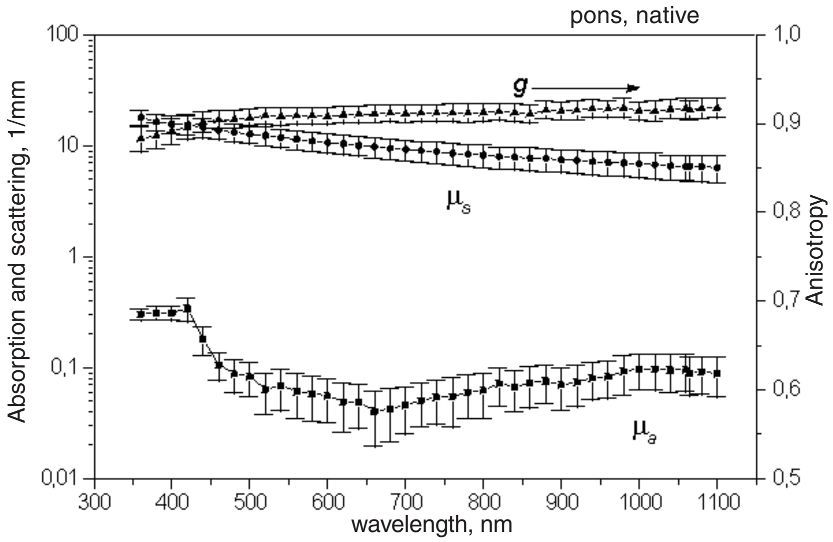
(a)



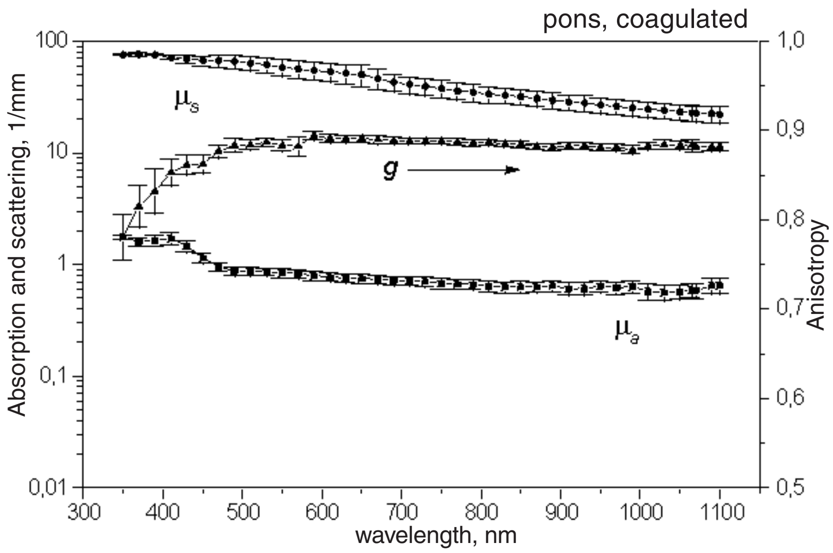
(b)

Figure 4. Optical properties of human cerebellum. Average of seven samples. Squares: absorption coefficient, circles: scattering coefficient, triangles: anisotropy factors and bars: standard errors. (a) Native samples and (b) coagulated samples.

matter (wm) were substantially higher than those of grey brain matter (gm) over the entire spectral range investigated (e.g. 800 nm: $\mu_t^{\mathrm{wm}} = 36.96\,\mathrm{mm}^{-1}$, figure 3(a); $\mu_t^{\mathrm{gm}} = 7.68\,\mathrm{mm}^{-1}$, figure 4(a)). The two brainstem tissues, namely pons and thalamus, also differed from each other. The scattering coefficients of pons were lower than those of thalamus for the complete wavelength range. Light penetration depth of diencephalic tissue, thalamus (see figure 6(a)), was greater than that of white matter but lower than that of grey matter. The tumour material



(a)

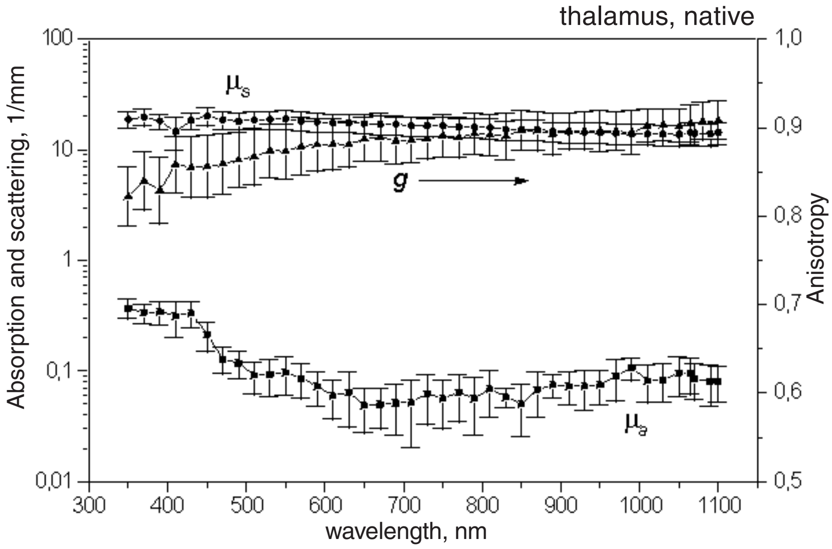


(b)

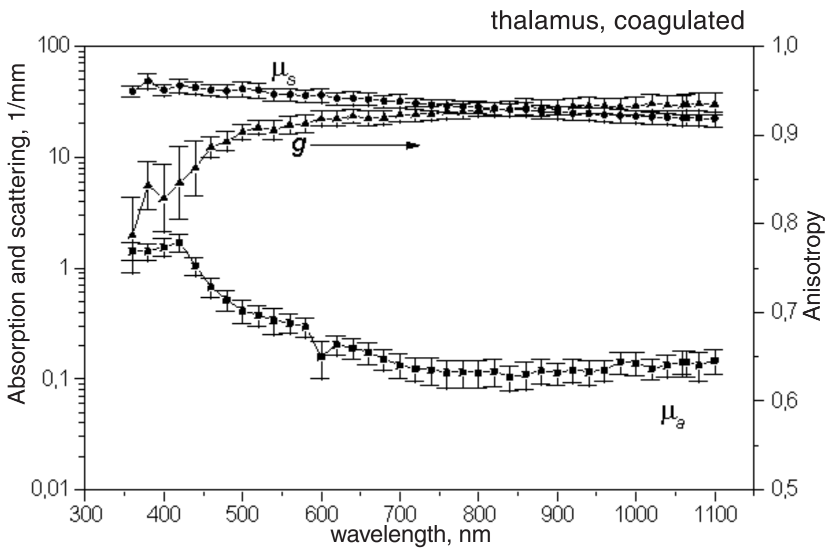
Figure 5. Optical properties of human pons. Average of seven samples. Squares: absorption coefficient, circles: scattering coefficient, triangles: anisotropy factors and bars: standard errors. (a) Native samples and (b) coagulated samples.

was generally macroscopically less homogeneous than any normal tissue investigated. The scattering coefficients and anisotropy factors of tumour tissues (figures 7(a) and (b)) were slightly higher than those of normal grey brain tissue. The astrocytoma tissue exhibited slightly higher penetration depths than the meningioma samples.

After coagulation, the values of absorption and scattering coefficients increased for all the investigated tissues. The extent of this increase, however, was different for each tissue



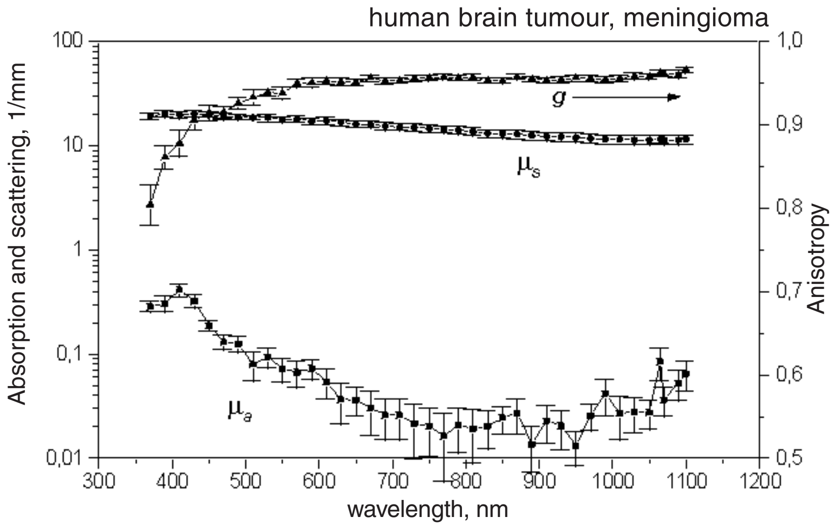
(a)



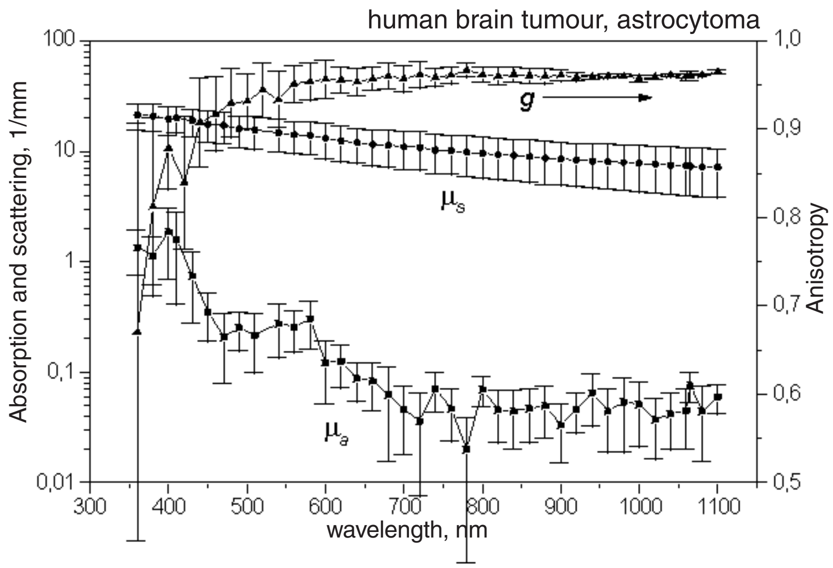
(b)

Figure 6. Optical properties of human thalamus. Average of seven samples. Squares: absorption coefficient, circles: scattering coefficient, triangles: anisotropy factors and bars: standard errors. (a) Native samples and (b) coagulated samples.

type. For white brain matter and for thalamus, the absorption coefficients increased by a factor of 2 to 3 and the scattering coefficients by a factor of 2. For grey brain matter, both the absorption and the scattering coefficients increased nearly threefold, whereas for pons both interaction coefficients were approximately four times higher. The absorption coefficients of cerebellum tissue increased by a factor of 3 to 5, whereas the scattering coefficients were found to be twice higher than those in untreated samples. The present study does not allow



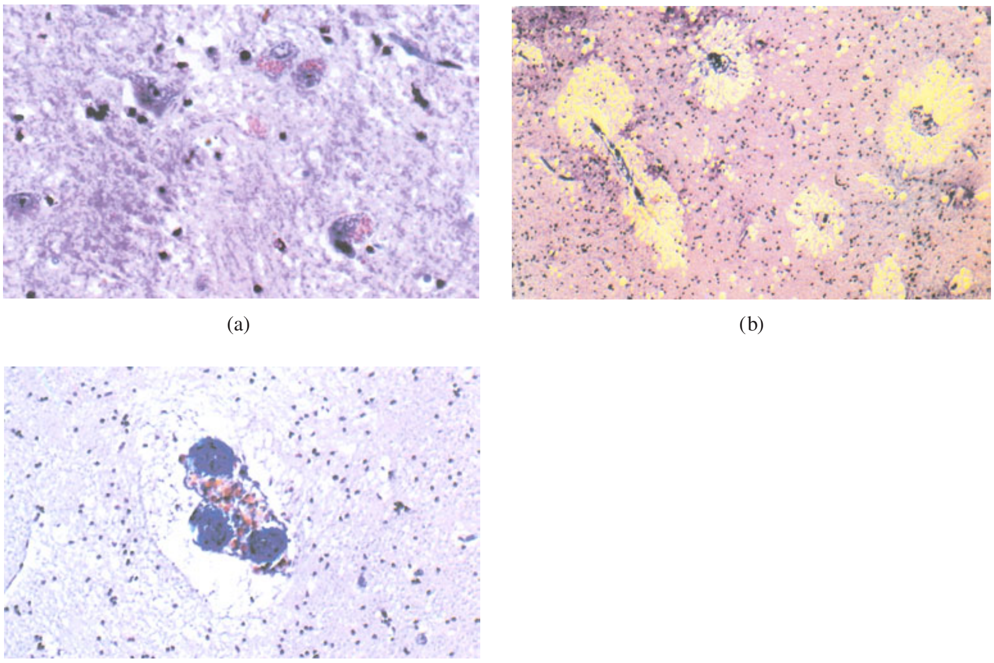
(a)



(b)

Figure 7. Optical properties of human brain tumours. Squares: absorption coefficient, circles: scattering coefficient, triangles: anisotropy factors and bars: standard errors. (a) Meningioma. Average of six samples. (b) Astrocytoma. Average of four samples.

quantitative explanation of the thermally induced changes of tissue optical properties, but it seems clear that such a significant increase of both interaction coefficients should be a result of substantial structural changes. Corresponding changes of human brain structures after coagulation were revealed by histological evaluation of the brain samples and compared to sections from non-coagulated tissue. Histological analysis has shown that coagulation causes considerable brain tissue shrinkage (figure 8(a)), condensation (figure 8(b)) as well as collagen swelling and homogenization of the vessel walls (figure 8(c)). Fine structure analysis of



(a) (b) (c)

Figure 8. Histological changes after brain tissue coagulation for 2 h at 80 °C. (a) Nuclear and cytoplasm condensation of neutropils and of nerve cells containing lipofuscin granules. (b) Shrinkage of white matter with extension of the Virchow–Robin perivascular spaces. (c) Nodular collagen swelling and homogenization of a blood vessel wall.

(This figure is in colour only in the electronic version)

heat-induced tissue alterations has been performed extensively over recent years on both *in vivo* and *ex vivo* models (Schober *et al* 1993, Schulze *et al* 2001). Our histological findings confirm the results obtained by Schober *et al* (1993) for rat brain tissue and by Schulze *et al* (1998) for laser coagulated pig brain tissues. It has been shown that specific phenomena, such as macroscopic shrinkage of brain tissue after coagulation, depend on the structural composition and are mainly determined by the loss of water by the brain tissue. As a result, concentration of chromophores and scattering inhomogeneities increases and tissues become optically denser, which leads to a significant increase of both scattering and absorption coefficients in the spectral range where water absorption is weak (up to 1100 nm–1300 nm). At the same time, tissue optical properties depend on the differences in the quantitative distribution of several histological substructures such as the cytoplasmatic compounds (typical for grey matter and thalamus with a high number of nerve cells) or axonal structures (typical for white matter with intercellular axonal connections). Moreover, after coagulation, these compounds are changed specifically in response to heat-induced protein denaturation and homogenization, thus changing all cellular structures according to their thermal vulnerability. These acute changes of tissue morphology and composition are likely to cause significant alterations in the microscopic distribution of the refractive index of brain tissue and result in corresponding changes of the macroscopic optical properties (i.e. absorption and scattering). It is probably worth noting that a similar increase of both absorption (by a factor of 2–10) and scattering (by a factor of 2–4) coefficients in the wavelength range from 500 nm to 1100 nm was recently reported for coagulated human blood by Black *et al* (2001).

Table 1. Optical properties of native human brain tissues, obtained in this work (∗) and known from the literature.

Biotissue	λ (nm)	μ_a (mm^{-1})	μ_s (mm^{-1})	g	μ_s' (mm^{-1})	δ_{eff} (mm)	Reference
White brain matter	456	0.81	92.3	0.92	7.384	0.22	Gottschalk (1992)
	450	0.14	42.0	0.78	9.24	0.5	∗
	514	0.5	104.5	0.93	7.315	0.29	Gottschalk (1992)
	510	0.1	42.6	0.81	8.094	0.64	∗
	630	0.15	38.6	0.86	5.404	0.63	Gottschalk (1992)
	630	0.08	40.9	0.84	6.544	0.79	∗
	675	0.07	43.6	0.87	5.668	0.83	Gottschalk (1992)
	670	0.07	40.1	0.85	6.015	0.83	∗
	850	0.08	14	0.95	0.7	2.3	Roggan *et al* (1994)
	850	0.1	34.2	0.88	4.1	0.9	∗
	1064	0.16	51.3	0.95	2.565	0.88	Gottschalk (1992)
	1064	0.04	11	0.95	0.55	3.76	Roggan *et al* (1994)
	1064	0.1	29.6	0.89	3.256	1.0	∗
Grey brain matter	456	0.9	68.6	0.95	3.43	0.29	Gottschalk (1992)
	450	0.07	11.7	0.88	1.404	1.84	∗
	514	1.17	57.8	0.97	1.734	0.31	Gottschalk (1992)
	510	0.04	10.6	0.88	1.272	2.52	∗
	630	0.14	47.3	0.93	3.311	0.83	Gottschalk (1992)
	630	0.02	9.0	0.89	0.99	4.06	∗
	675	0.06	36.4	0.91	3.276	1.29	Gottschalk (1992)
	670	0.02	8.4	0.9	0.84	4.4	∗
	1064	0.19	26.7	0.96	1.07	1.18	Gottschalk (1992)
	1064	0.05	5.7	0.9	0.57	3.28	∗

Table 2. Optical properties of coagulated human brain tissues, obtained in this work (∗) and known from the literature.

Biotissue	λ (nm)	μ_a (mm^{-1})	μ_s (mm^{-1})	g	μ_s' (mm^{-1})	δ_{eff} (mm)	Reference
White brain matter	850	0.09	17	0.94	1.02	1.83	Roggan *et al* (1994)
	850	0.09	30	0.88	3.6	1.01	∗
	1064	0.05	13	0.93	0.91	2.6	Roggan *et al* (1994)
	1064	0.1	27	0.89	2.97	1.06	∗

Details of the thermally induced changes of tissue structure depended on the tissue type. This statement may be confirmed by the absence of common tendency in the thermal changes of the anisotropy factors. As demonstrated in figures 2(a) and (b), and 3(a) and (b), the anisotropy factors of white brain matter and grey brain matter remained nearly unchanged. The anisotropy factors of cerebellum decreased in the visible spectral range and did not change in the infrared (figures 4(a) and (b)). The anisotropy factors of pons tissue slightly decreased (figures 5(a) and (b)) whereas the anisotropy factors of thalamus were found to be slightly increased (figures 6(a) and (b)).

Published data on the optical properties of white and grey brain matter are rare. The optical properties of brainstem tissue (pons, thalamus), cerebellum and brain tumours have not been investigated before. In tables 1 and 2 the optical properties of white and grey brain matters published in the literature so far are compared to those obtained in this work. The values of absorption coefficients, scattering coefficients, anisotropy factors, reduced scattering coefficients and light penetration depths were evaluated and presented in the tables. Gottschalk (1992) has evaluated the optical properties of native white and grey brain matters for a wide spectral range from integrating-sphere measurements using an inverse δ-Eddington method. Comparing the results of the present work with these findings it is obvious that both studies found the light penetration depth of white brain matter to be substantially lower than that of grey matter. On the other hand, we obtained in general lower values for absorption and scattering coefficients. This discrepancy may be explained by the different theoretical approach. The δ-Eddington model that Gottschalk (1992) employed to extract the optical properties from the integrating-sphere measurements is one dimensional. Therefore, he could not account for the losses of light at the side edges of the samples. This might have led to an overestimation of the extinction coefficients. Roggan *et al* (1994) have determined the optical properties of native and coagulated homogenized white brain matter at the wavelengths of 850 nm and 1064 nm using a double-integrating-sphere technique combined with an inverse Monte Carlo method. The results of Roggan *et al* (1994) qualitatively confirm the tendency of the interaction coefficients to decrease while the wavelength increases. The quantitative comparison of the results obtained in these two studies shows that the extinction coefficients of native white matter, determined by Roggan *et al* (1994), are lower than the values obtained in the present work. These discrepancies might be explained by the different sample preparation techniques. In the study of Roggan *et al* (1994), the tissues were subjected to shock freezing and homogenization. This procedure may lead to destruction of the natural sample inhomogeneities and consequently to lower extinction coefficients of the sample. Indeed, a later paper of the same group (Roggan *et al* 1999) demonstrated the scattering coefficient of liver samples to be lowered by cryo-homogenization by 29% in comparison to the untreated ones. On the other hand, Roggan *et al* (1999) have shown that the storage procedure leads to considerable increase of the scattering coefficient values in liver samples. Therefore, the storage of brain tissues at low temperatures could have increased the values of scattering coefficients obtained in the present study. However, it should also be mentioned that liver tissue has a high content of enzymes and, therefore, is subject to a faster post-mortem proteolysis than brain tissue.

There are very few data published on the optical properties of coagulated brain tissue. Table 2 shows the optical properties of coagulated human white brain matter published by Roggan *et al* (1994) in comparison with those obtained in the present study. Both studies demonstrated a substantially reduced light penetration depth after coagulation. The quantitative differences between the obtained data may be again explained by differences in sample preparation techniques.

One of the most important questions of tissue optics is whether the *in vitro* optical properties are close enough to the optical properties of living tissues to be used for *in vivo* applications. Without a doubt, tissue properties change after the death of the organism. There is, however, evidence that for many biomedical applications the optical properties determined *in vitro* can provide information adequate for *in vivo* use. For example, the optical properties of human brain tissues obtained in present study can be used to plan laser-induced interstitial thermotherapy. As was shown (Yaroslavsky *et al* 1996a, Schwarzmaier *et al* 1998), the results of the planning procedure based on the respective *in vitro* data were in good agreement with the final lesion sizes induced in the patients' brain after laser irradiation as determined from NMR images. Discrepancies between the treatment simulation and the final lesion size in the patients were shown to be most probably due to perfusion inhomogeneities. Thus, the optical data obtained *in vitro* using the combination of integrating-sphere measurements and the inverse Monte Carlo technique are obviously useful for *in vivo* applications.

4. Conclusions

The analysis of the results obtained in this study shows that our data are in qualitative agreement with the values reported in the literature when the latter are available. The discrepancies as well as the variations within the values reported in the literature for selected brain tissue structures are most probably due to the different theoretical models and sample preparation techniques employed.

Our results clearly demonstrate that the optimal wavelength range for therapeutic and diagnostic light applications lies in the range of 1000–1100 nm, where the light penetration depth in the visible and near infrared region is maximal. This conclusion is based on the following facts. On the one hand, scattering coefficients of all the structures investigated decrease with increasing wavelength. On the other, the absorption coefficient is quite low in this spectral range, since the oxyhaemoglobin absorption decreases and water absorption is still weak.

The optical properties of all measured brain tissues changed dramatically after coagulation. This is an important factor, which should be carefully considered in treatment planning, since these changes can significantly influence the resulting distribution of the optical radiation within tissue, and consequently the outcome of therapeutic procedure. During the coagulation process, the effective penetration depth of light into the tissue decreases for all the tissues investigated. Thus if laser irradiation leads to the structural changes within the tissue, the changes in the light penetration depth should be accounted for while planning the therapeutic procedure.

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